

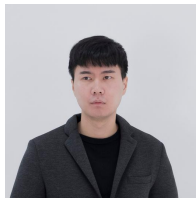
Functional Connectivity Reconfiguration in N-back Working-Memory

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Meet our team!



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Does load-related brain connectivity predict working-memory performance?

We predicted two things:

Pattern

A 78-feature FC fingerprint of the 2-back - 0-back change predicts performance.

Distributed pattern

Direction

Higher load shifts networks toward integration; larger shifts accompany better performance.

One-number direction

A distributed pattern and a one-number direction are different claims

We evaluated a 78-feature FC difference in held-out people

FC reconfiguration = how much each network pair changes its coupling when memory load goes from 0-back to 2-back.

1. Cohort

336 participants
HCP N-back dataset
2 WM runs per subject
360 Glasser ROIs
TR = 0.72 s

2. Condition frames

4 s HRF shift window
312 frames per load
No 0-back / 2-back temporal overlap

3. FC per condition

360 × 360 Pearson correlation matrix
Computed separately for 0-back and 2-back

4. Network fingerprint

78 total values over
Cole-Anticevic networks:
• 12 within-network
• 66 between-network

5. Reconfiguration

fingerprint(2-back) - fingerprint(0-back)
Resulting feature matrix:
336 participants × 78 features

6. Model pipeline

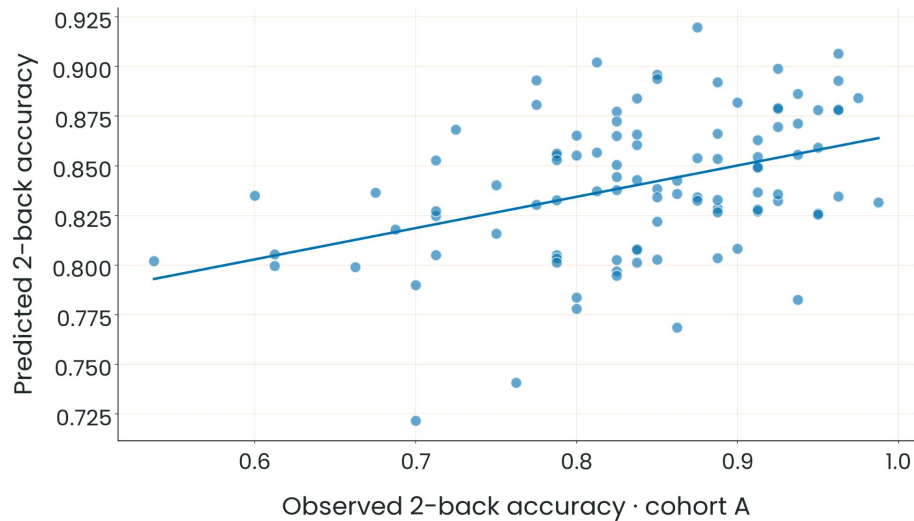
StandardScaler + RidgeCV
Cross-validated regression model.
Fitted strictly inside each training fold.

7. Held-out evaluation

Repeated 5-fold CV • Fixed holdout partition • 1000-permutation null distribution • **B → A transfer** (301 → 100, identity-disjoint)
All splits hold out participants; “±” represents split SD, not CI.



The FC pattern predicted performance and transferred to a separate cohort



B → A Transfer Result

$r = 0.398$

Bootstrap 95% CI [0.25, 0.53]

301 B-only → 100 A (35 shared identities removed)

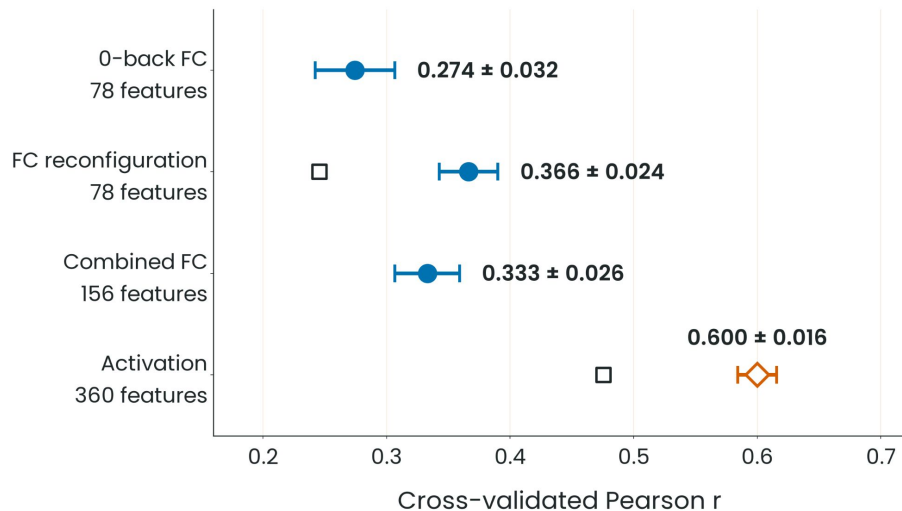
Primary Repeated CV

$r = 0.366 \pm 0.024$

336 participants · 78 FC features

- Identity-disjoint same-HCP transfer—not independent-site validation.
- SD summarizes sensitivity across 20 overlapping partitions, not population uncertainty.

The prediction held—but a simpler regional signal predicted better



Repeated-CV Correlation (mean ± split SD)

0-back FC: **0.274 ± 0.032**

FC reconfiguration: **0.366 ± 0.024**

0-back + reconfiguration: **0.333 ± 0.026**

Activation contrast: **0.600 ± 0.016**

Direction, as predicted—but only at group level

Segregation fell under load ($0.3271 \rightarrow 0.3035$; $\Delta = -0.0236$; $p = 3.45 \times 10^{-5}$), yet larger shifts did not predict better performance ($r = -0.105$; $p = .054$).

Methodological Caveats & Guardrails

- **No biological winner claimed:** Unmatched features (360 activation vs 78 network FC). Activation does not erase FC findings but challenges specificity.
- **Not load-specific:** 0-back activation alone predicts as well (0.571), indicating a general benchmark.
- **Controls:** Activation is mean BOLD difference (not GLM beta). No cerebrovascular reactivity (CVR) control exists.
- **Open squares:** Held-out cross-run generalization.

Predictive signal survives; connectivity-specific mechanism remains unresolved

Survives

The pattern hypothesis

- **A 78-feature FC difference** predicts unseen 2-back accuracy.
- **The model transfers** across identity-disjoint same-HCP cohorts.

Refined

The directional hypothesis

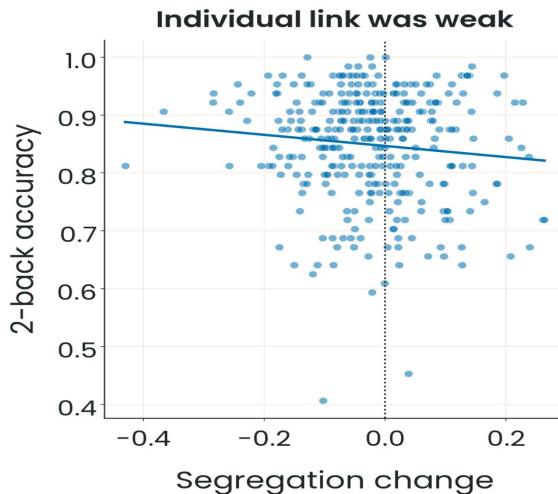
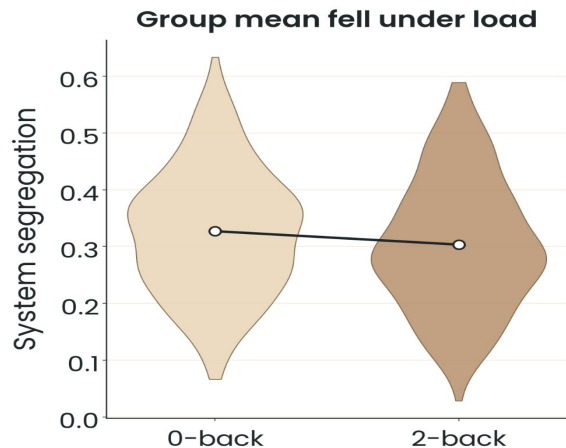
- **Segregation fell** under load, but larger shifts did not predict better performance.
- **Reconfiguration** showed no clear gain beyond 0-back FC.
- **FC added no clear gain** over activation under the current unmatched comparison.

Unresolved

Mechanistic open questions

- **Is the predictive info** connectivity-specific, or shared with task activation?
- **Vascular reactivity (CVR)** is not controlled in the activation benchmark.

Group direction was real; the individual link was weak



Statistical Summary

Group mean: 0-back 0.3271 $\rightarrow$ 2-back 0.3035

Paired change: $\Delta = -0.0236$ ($p = 3.45 \times 10^{-5}$)

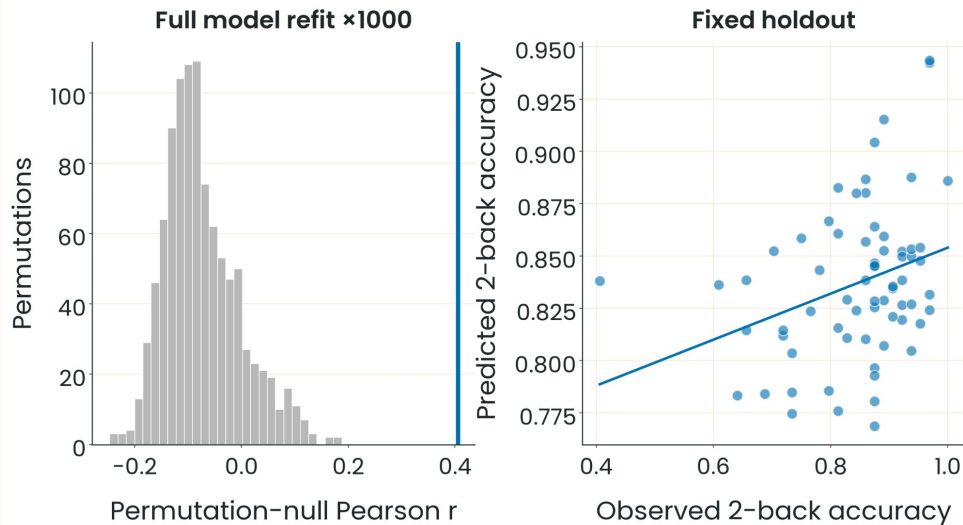
Across participants: $r = -0.105$ ($p = .054$)

Key Takeaways & Caveats

A reliable group-level mean shift does not establish individual predictive relevance.

Methodological limit: System segregation is a Chan-style scalar summary, not a replacement for the 78-feature multivariate pattern.

The checks behind the primary result



Validation Details

Fixed holdout: $r = 0.312$ in 67 unseen participants

Full-refit null (seed 42): $r = 0.405$; $p = 1/1001 \approx .001$

B→A A-label permutation: $p = 1/1001 \approx .001$

Reconfiguration over 0-back FC: $\Delta R^2 = +0.0344 \pm 0.0225 \rightarrow$ no clear gain

FC over activation: $\Delta R^2 = -0.0030 \pm 0.0065 \rightarrow$ no clear gain

Methodological Notes & Caveats:

The permutation p -value belongs only to seed-42 $r = 0.405$; the holdout is a separate split. The 2-SD decision rule is a split-sensitivity heuristic, not a formal superiority or equivalence test.

Four tests could resolve the remaining question

01

Match activation and FC dimensionality in nested participant-level CV.

02

Separate task coactivation from coupling with a prespecified estimator.

03

Test independent-site, repeat-session, family-aware generalization.

04

Run the activation model on resting-state amplitude: if rest predicts 2-back accuracy as well as task frames, the signal is task-independent.

Decision Criterion

FC must add reliable held-out value beyond activation and single-condition FC.

Note: These represent proposed validation experiments rather than completed research pipeline analyses.

References and statistical guardrails

Key Literature & References

Avery et al. (2020)

Distributed patterns of FC predict WM · J. Cognitive Neuroscience · PMC8004893

Chan et al. (2014)

Decreased segregation of brain systems · PNAS

Murphy et al. (2020)

Multimodal network dynamics underpinning WM · Nature Communications · 10.1038/s41467-020-15541-0

Masharipov et al. (2024)

Task-modulated FC methods · Communications Biology · 10.1038/s42003-024-07088-3

Hedge et al. (2018)

The reliability paradox · Behavior Research Methods · 10.3758/s13428-017-0935-1

Logothetis (2008)

What we can do and what we cannot do with fMRI · Nature

Full annotated bibliography: [manuscript/references.md](#)

Statistical Guardrails & Definitions

$$r = 0.366 \pm 0.024$$

Mean $\pm$ split SD across 20 repeated-CV partitions.

$$\text{seed-42 } r = 0.405; p \approx .001 (1/1001)$$

Full-refit permutation baseline.

$$B \rightarrow A \text{ } r = 0.398; 95\% \text{ CI } [0.25, 0.53]$$

Identity-disjoint same-HCP transfer metric.

Scientific Caveat: References frame and contextualize limits; they do not establish exact replications or causal validation. Logothetis (2008) frames the CVR vascular reactivity gap as an external cited limitation rather than our own direct finding.



INTERNAL SLIDE: Compression map

15-Slide Internal-Review Version → Final Presentation Deck

Spoken Slides (Deck 1)

Slide 3: Absorbs Question and both hypotheses into one opening beat.

Slide 4: Cohorts, feature construction, and evaluation mapped to a pipeline schematic.

Slide 5: Focuses on Primary CV and transfer (sub-checks demoted to backup Slide 9).

Slide 6: Directional refinement & activation benchmark (segregation figure demoted to backup 8).

Slide 7: Conclusion, explicitly naming both opening hypotheses.

Backup & Omitted Assets

Backup Slide 8: Directional figure and full statistics.

Backup Slide 9: Null, fixed holdout, and incremental test.

Backup Slide 10: Future work, expanded to four tests.

Omitted Assets:

- Internal Slide 10 (anatomical context) stays out of both decks.
- Several assets remain unassigned in the chart library: condition-fc-contrast, anatomical-context, feature-construction, primary-repeated-cv, and incremental-fc-test.